SAT Hard Question
Set 12

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Phase 2 · Hard Question Practice

Overview

Question 1

In the given equation, a and b are non-zero constants. Which of the following expressions gives a solution to the equation?

$$7ax^2 + (7b - 12a)x - 12b = 0$$

A. $-\frac{a}{b}$

B. $-\frac{b}{a}$

C. $\frac{a}{b}$

D. $\frac{b}{a}$

Answer 1

Correct Answer: B

Question 2

In the given equation, a and b are constants. The equation has infinitely many solutions. What is the value of b?

$$\frac{ax+b}{3} = \frac{6}{17}x - \frac{13}{12}$$

A. $-\frac{18}{17}$

B. $\frac{18}{17}$

C. $-\frac{13}{4}$

D. $\frac{13}{4}$

Answer 2

Correct Answer: C

Question 3

In the xy-plane, a parabola has x-intercepts at $(-17, 0)$ and $(37, 0)$. If the equation of the parabola is written as:

$$y = ax^2 + bx + c,$$

where a, b, and c are constants, which of the following expressions gives c in terms of a?

A. $c = -1080a$

B. $c = -629a$

C. $c = \frac{a}{34}$

D. $c = 51a$

Answer 3

Correct Answer: B

Question 4

In triangle ABC, the measure of angle B is 90° and BD is an altitude of the triangle. The length of AB is 15, and the length of AC is 23 greater than the length of AB. What is the value of $\frac{BC}{BD}$?

A. $\frac{15}{38}$

B. $\frac{15}{23}$

C. $\frac{23}{15}$

D. $\frac{38}{15}$

Answer 4

Correct Answer: D

Question 5

A certain investment account offers a special interest rate for the first 4 months the account is open followed by a lower interest rate for the remainder of the time the account is open. Akeyla opened one of these accounts with an original account balance of $700.00 and did not make any other deposits or withdrawals.

- 4 months after Akeyla opened the account, the balance had increased by 0.6% of the original balance.
- 6 months after opening, the balance had increased by an additional 0.1% of the balance at the end of the first 4 months.
- Every 2 months after the first 6 months, the balance increased by an additional 0.1% of the balance 2 months before.

Which of the following equations could represent the account balance $B(x)$, in dollars, x months after the account was opened, where $x \geq 4$?

A. $B(x) = 704.20(1.001)^{\frac{x-4}{2}}$

B. $B(x) = 700.00(1.001)^{\frac{x-4}{2}}$

C. $B(x) = 704.20(1.001)^{2(x-4)}$

D. $B(x) = 700.00(1.001)^{2(x-4)}$

Answer 5

Correct Answer: A

Question 6

The area of a rectangular region is increasing at a rate of 180 square centimeters per hour. What is this rate, in square meters per minute?

A. 3

B. 0.3

C. 0.03

D. 0.0003

Answer 6

Correct Answer: D

Question 7

The exponential function $f(x)$ is defined by

$$f(x) = ab^x$$

where a and b are positive constants.
If

$$f(n-1) = f(n) + \frac{82}{100} f(n-1)$$

where n is a constant, what is the value of b?

A. 1

B. 0.82

C. 0.18

D. 10

Answer 7

Correct Answer: C

Question 8

The function f is defined by the given equation, where k is a constant. The value of $f(x)$ increases by 83% for every increase of x by 1.

$$f(x) = k(1.83)^x$$

For which of the following functions, where k is a constant, does the value of $g(z)$ increase by 83% for every increase of z by $\frac{1}{4}$?

A. $g(z) = k(1.83^z)^{\frac{1}{4}}$

B. $g(z) = k(1.83)^{z+\frac{1}{4}}$

C. $g(z) = k(1.83^z)^4$

D. $g(z) = k(1.83)^{z-4}$

Answer 8

Correct Answer: C

Question 9

A right square pyramid has a height of 11 centimeters (cm). A second right square pyramid has a height of 22 centimeters. The area of the bases of each of the two pyramids is 100 cm^2.
Which of the following is closest to the difference in the surface area of the second pyramid and the surface area of the first pyramid, in cm^2?

A. 209
B. 220
C. 367
D. 551

Answer 9

Correct Answer: A

Thank You!

We appreciate your time and focus.

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